

Logistic Regression for Probability-Based Insights

Agenda

- Introduction to Logistic Regression
- Understanding the Confusion Matrix
- Classification Metrics and Optimal threshold
- Statistical Inference for Logistic Regression
- ROC Curve

Let's begin the discussion by answering a few questions

Logistic Regression Quiz

You are building a model where the dependent variable indicates whether a customer will churn (1) or not (0). Which reason best explains why Logistic Regression is more suitable than Linear Regression?

A

Logistic Regression gives a straight-line fit which is better for binary outputs.

B

Logistic Regression restricts predictions between 0 and 1, making them interpretable as probabilities.

C

Linear Regression gives probabilities that always stay between 0 and 1

D

Logistic Regression always gives higher accuracy than Linear Regression.

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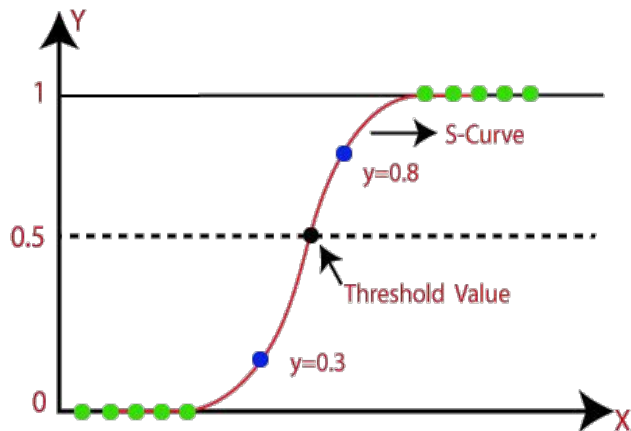
D

Logistic Regression always gives higher accuracy than Linear Regression.

Introduction to Logistic Regression

Logistic Regression is a supervised learning algorithm that estimates the log of odds for an event which can be used to predict the probability of the occurring of that event.

Logistic Regression is used when the outcome variable is categorical or discrete, such as yes/no or multiple classes. It estimates the probability of an event occurring based on one or more independent variables.



The model uses the logistic (sigmoid) function to map predicted values to a range between 0 and 1. The predicted probabilities can be converted to classes based on the threshold (the default threshold is 0.5).

Logistic Regression Quiz

A medical diagnostic model has high Accuracy but very low Recall. What scenario does this indicate?

A

The model is incorrectly labelling many actual positives as negatives.

B

The model is correctly identifying most actual positive cases.

C

The model is making too many false positive predictions.

D

The model correctly classifies all negatives but misclassified few positives.

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Confusion Matrix

A **tabular representation** of the **predicted vs actual classes** and it can be used to assess the **performance of a classification model**

We have four possible outcomes from a classification model

	Predicted Positive	Predicted Negative
Actual Positive	TP	FN
Actual Negative	FP	TN

Confusion Matrix

True Positive (TP)

**Actual Positive
Predicted Positive**

Case 1

The application was approved and the model also predicted that it would be approved

False Negative (FN)

**Actual Positive
Predicted Negative**

Case 2

The application was approved but the model predicted that it wouldn't be approved

False Positive (FP)

**Actual Negative
Predicted Positive**

Case 3

The application wasn't approved but the model predicted that it would be approved

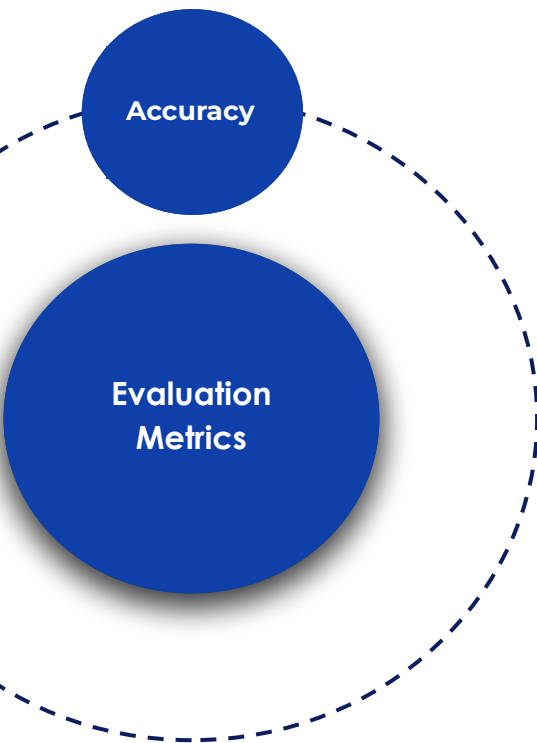
True Negative (TN)

**Actual Negative
Predicted Negative**

Case 4

The application wasn't approved and the model also predicted that it wouldn't be approved

Classification Metrics

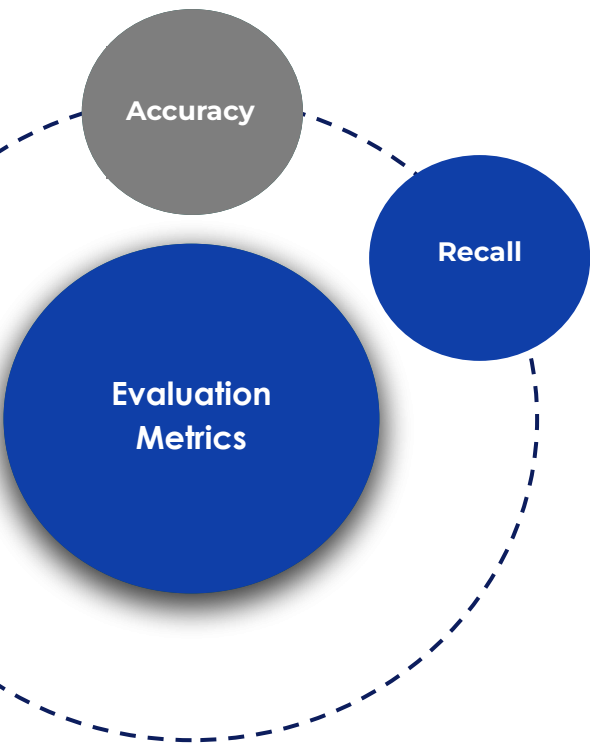


- Most intuitive and widely used metric

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

- Represents the **proportion of correct predictions** out of the total number of predictions
- Ranges from 0 to 1

Classification Metrics



- The ratio of true positives to total actual positives

$$Recall = \frac{TP}{TP + FN}$$

- Tells us out of all the 'positive' instances, how many did the model remember correctly
- Ranges from 0 to 1

Recall ↑ False Negatives ↓

Logistic Regression Quiz

A hospital uses a logistic regression model to predict high-risk stroke patients (1), and the test results show 92% accuracy, 60% precision, and 40% recall. Based on this scenario, which metric should the hospital prioritize for screening, and what does this imply about the model's suitability?

A

Metric: Accuracy, Approve Current Model

C

Metric Precision : Approve Current Model

B

Metric: Recall, Reject Current Model

D

Metric: Accuracy, Reject Current Model

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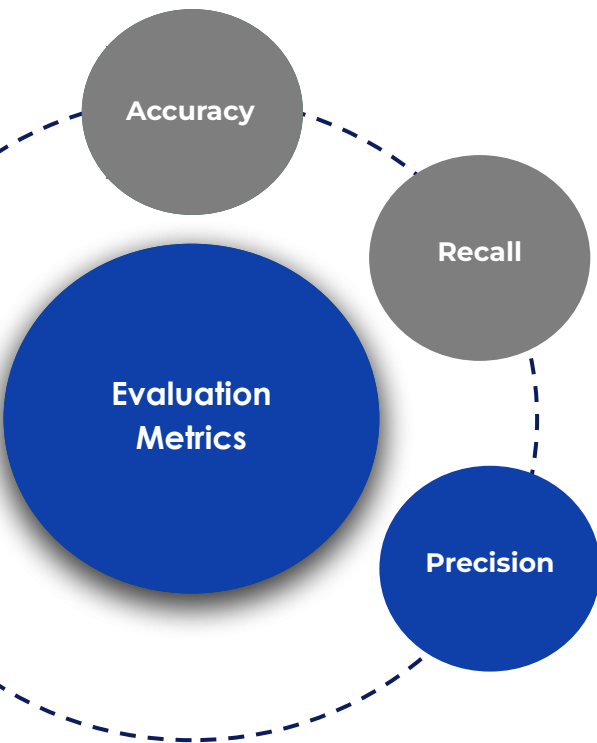
B

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D

Metric: Accuracy, Reject Current Model

Classification Metrics



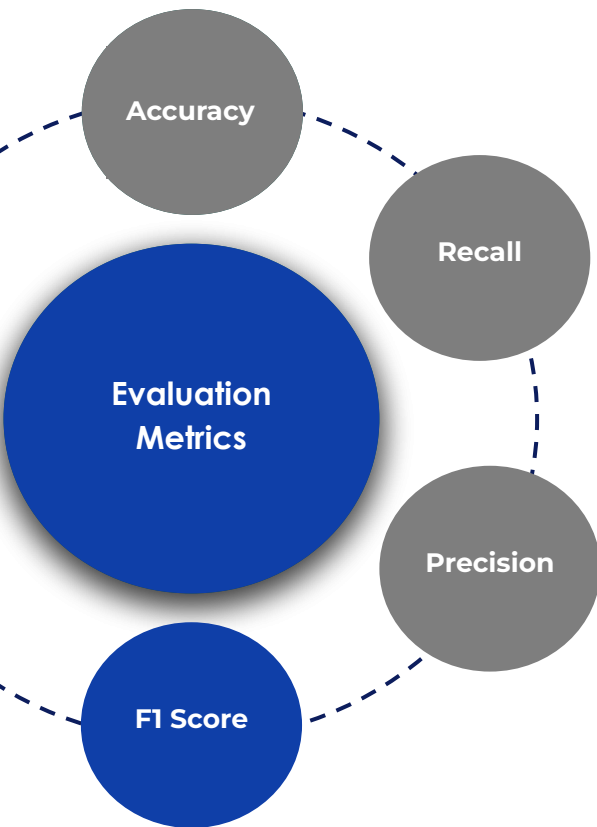
- The ratio of true positives to total predicted positives

$$Precision = \frac{TP}{TP + FP}$$

- Tells us out of all the predicted 'positive' instances, how many are actually 'positive'
- Ranges from 0 to 1

Precision ↑ False Positives ↓

Classification Metrics



- Maintains a balance between False Positives and False Negatives

$$F1\ Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

- Ranges from 0 to 1

F1 Score  False Positives  False Negatives 

Logistic Regression Quiz

A logistic regression model is built to determine the likelihood of diabetes based on patient demographic and health attributes, The coefficient of the Age variable is 0.0122.

Which of the following is a valid statistical inference?

A

For every unit increase in the Age variable, the output increases by 0.0122, keeping all other variables constant.

B

For every unit increase in the Age variable, the log-odds of the output increases by 0.0122, keeping all other variables constant.

C

For every unit increase in the Age variable, the output remains unchanged.

D

For every unit increase in the Age variable, the output changes by a random value.

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Statistical Inference from a Logistic Regression Model

The **coefficients of the logistic regression** model are in terms of **log-odds**

For every one-unit increase in **Age**, the log-odds of the outcome increases by **0.0122**, holding other variables constant

A one-unit increase in **Blood Pressure** decreases the log-odds of the outcome by **0.0038**, all other variables held constant

The **interpretation of p-values** in logistic regression is **similar to that in linear regression**.

Logit Regression Results

Dep. Variable:	Class	No. Observations:	537			
Model:	Logit	Df Residuals:	528			
Method:	MLE	Df Model:	8			
Date:	Mon, 03 Mar 2025	Pseudo R-squ.:	0.3045			
Time:	20:04:29	Log-Likelihood:	-241.39			
converged:	True	LL-Null:	-347.09			
Covariance Type:	nonrobust	LLR p-value:	2.518e-41			
=====						
	coef	std err	z	P> z	[0.025	0.975]

const	-9.7226	1.021	-9.522	0.000	-11.724	-7.721
Pregnancies	0.1445	0.039	3.706	0.000	0.068	0.221
Glucose	0.0363	0.005	7.819	0.000	0.027	0.045
BloodPressure	-0.0038	0.010	-0.359	0.720	-0.024	0.017
SkinThickness	0.0008	0.016	0.048	0.962	-0.031	0.032
Insulin	0.0010	0.002	0.644	0.519	-0.002	0.004
BMI	0.0936	0.022	4.186	0.000	0.050	0.137
Pedigree	1.0345	0.351	2.950	0.003	0.347	1.722
Age	0.0122	0.011	1.070	0.284	-0.010	0.035
=====						

Logistic Regression Quiz

If you increase the classification threshold from 0.5 to 0.7 in a binary classification model, which change is most likely?

A

Recall will increase, and Precision will decrease.

B

Both Recall and Precision will increase.

C

Precision will increase, but Recall will decrease.

D

Both Recall and Precision will decrease.

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Choosing the Optimal Classification Threshold

The **default threshold** in a logistic regression model is **0.5**, i.e., if the probability of being positive is **greater than 0.5**, the sample is classified as **positive (1)**, **else** it is classified as **negative (0)**.

However, this **threshold may not be appropriate for all use cases**.

For example, in a **medical diagnosis scenario**, a **false negative** (classifying a sick patient as healthy) is **more severe than a false positive** (classifying a healthy patient as sick).

In such a case, we generally **decrease the threshold value to reduce false negatives**.

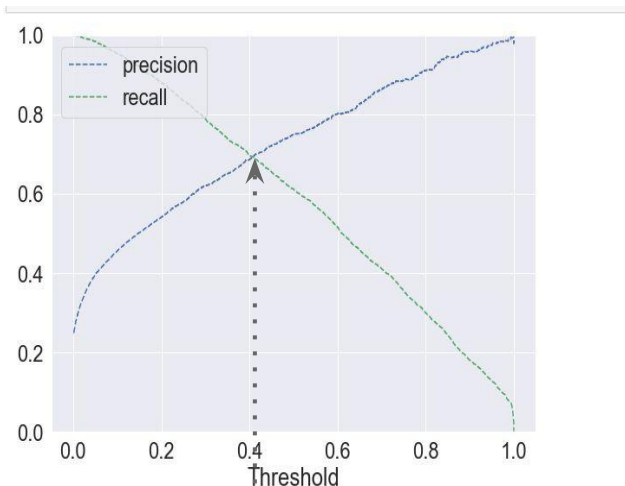
Defining an **optimal threshold value is crucial** in controlling the performance of the model.

Choosing the Optimal Classification Threshold

Precision–Recall is valuable for assessing model performance on imbalanced datasets by showing how both metrics change with different thresholds. This can be used to select optimal threshold as required to improve the model improvement

Here we can see, Precision and Recall are the same when the threshold is 0.4. Increasing the threshold enhances Precision, while lowering it improves Recall.

Choosing a threshold can completely change the model performance assessment; it is important to identify the 'sweet spot'.



Logistic Regression Quiz

Based on the ROC curves shown in the plot, which threshold demonstrates the best performance for a classification problem?

A

Threshold 1

B

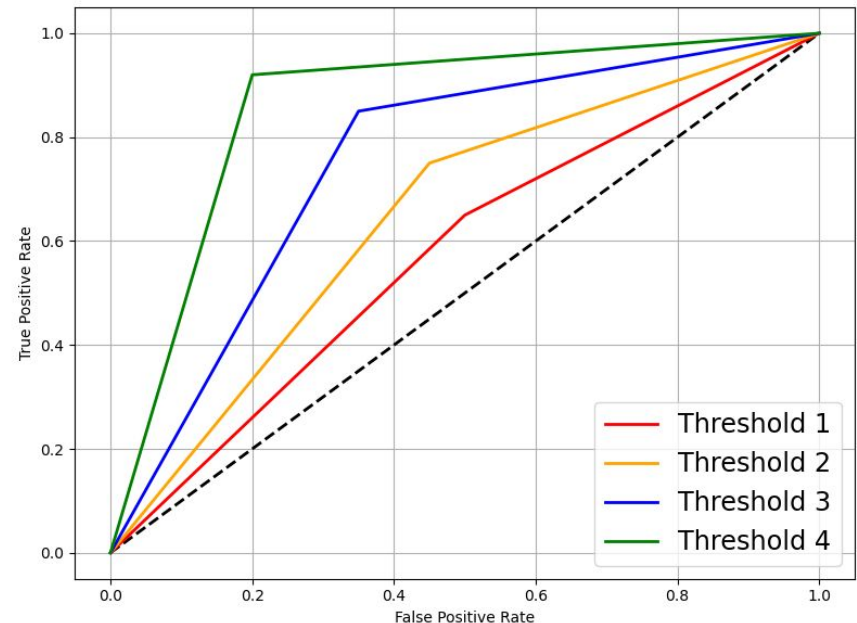
Threshold 2

C

Threshold 3

D

Threshold 4



Logistic Regression Quiz

Based on the ROC curves shown in the plot, which threshold demonstrates the best performance for a classification problem?

A

Threshold 1

B

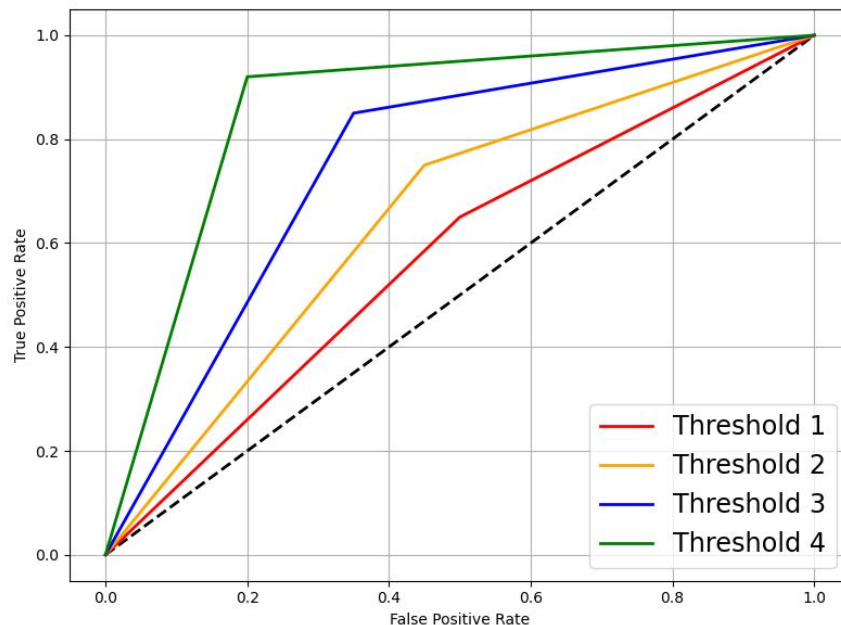
Threshold 2

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Threshold 3

D

Threshold 4



ROC (Receiver Operating Characteristic) Curve

Different threshold values cause the **TPR** and **FPR** values to **change**, which in turn gives different **AUC** values.

The threshold for which we get the **highest AUC** will be the **optimal threshold** for that classifier

The value of AUC ranges from 0 to 1

0

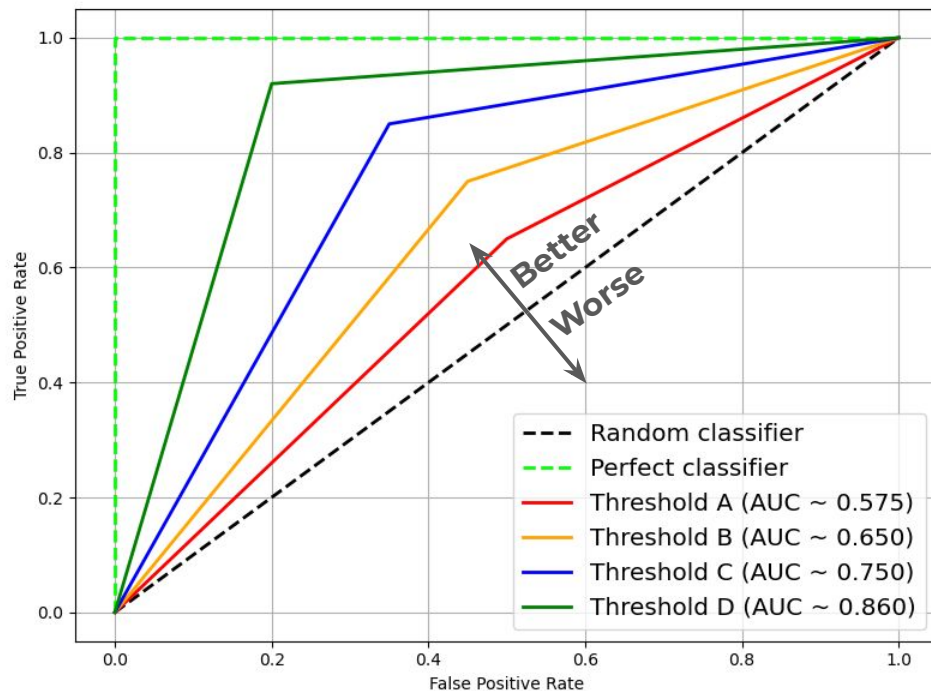
The worst classifier: All instances incorrectly classified

0.5

A random classifier with no discrimination for positive or negative classes

1

A perfect classifier: All instances correctly classified





Power Ahead !



Summary: Evaluation Metrics for Classification

Accuracy	Precision	Recall	F1-Score
$\frac{TP + TN}{TP + TN + FP + FN}$	$\frac{TP}{TP + FP}$	$\frac{TP}{TP + FN}$	$2 \times \frac{Precision \times Recall}{Precision + Recall}$
Represents the proportion of correct predictions out of the total number of predictions	The ratio of true positives to total predicted positives	The ratio of true positives to total actual positives	Most credible metric to weigh the model performance even if the class distribution is imbalanced
Suitable for balanced datasets only	Suitable for both balanced and imbalanced datasets	Suitable for both balanced and imbalanced datasets	Suitable for both balanced and imbalanced datasets
Ranges from 0 to 1	Ranges from 0 to 1	Ranges from 0 to 1	Ranges from 0 to 1
Increase in Accuracy reduces both False Positives and False Negatives. However, it can be misleading in case of imbalanced datasets	Increase in Precision reduces False Positives and vice-versa	Increase in Recall reduces False Negatives and vice-versa	Increase in F1-Score reduces both False Positives and False Negatives, even if the dataset is imbalanced